Planning to Promote Coastal Sustainability:

Mitigative Measures for Communities Impacted by Climate Change

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The use of nonrenewable resources directly contributes to the proliferation of climate change and subsequent hazardous weather conditions. These exacerbated weather conditions can impact individuals, and other beings, in all walks of life, especially those living along the coast. In order to mitigate the impacts felt by said coastal communities, it is essential to develop adaptive planning measures that consider the Earth's dynamic climate.

Society's Reliance on Nonrenewable Resources

Nonrenewable resources are resources that "cannot be replaced" once they are used (National Geographic Society, 2023). These resources take numerous forms, including "coal, natural gas, [and] oil," and are collectively referred to as "fossil fuels" (National Geographic Society, 2023). Fossil fuels are composed of biological matter, which has been compressed and heated within the Earth's subsurface for millions of years (National Geographic Society, 2023). Once fossil fuels are extracted from the underground, they can be burned to generate energy for a variety of purposes: gas-dependent transportation, industrial processes, etc. For reference, as of 2023, "approximately 80 percent" of total global energy utilized was generated from nonrenewable sources alone (National Geographic Society, 2023). Society's reliance on nonrenewable resources can primarily be attributed to the existence of pre-existing infrastructure (factories, automobiles, etc.), high energy output, and general affordability. However, these nonrenewable energy sources are severely unsustainable, as they directly contribute to the proliferation of the Earth's changing climate.

Nonrenewable's Contribution to Global Climate Change

As previously mentioned, fossil fuels are burned in order to generate energy for a variety of purposes. During this burning stage, fossil fuels can emit an abundance of greenhouse gasses (GHG). GHGs are atmospheric gasses, such as carbon, which "trap the sun's heat" within the Earth's upper atmosphere (United Nations, n.d.-a). For reference, GHGs naturally accumulate within the Earth's atmosphere as natural processes occur. For example, carbon is released when living species respirate or when fires ignite and burn matter. These naturally accumulated GHGs then trap the sun's heat energy, acting as a blanket to maintain the Earth's average temperatures. However, as society excessively burns nonrenewable resources, the Earth's natural GHG balance is offset, leading to excessive heat capture. For this, and many other reasons, "the world is now warming faster than at any point in recorded history" (United Nations, n.d.-a). These intensifying climate conditions have subsequently exacerbated global weather patterns, resulting in the disruption of various social and ecological communities.

Climate Change's Contribution to Altered Weather Conditions

Similar to the upper atmosphere, the Earth's oceans directly accumulate heat-trapping GHGs. According to the National Oceanic and Atmospheric Administration (NOAA), "the ocean acts as a 'carbon sink' and absorbs about 31% of the CO2 emissions released into the atmosphere" (2024). As societies across the globe continue to excessively burn nonrenewables, the quantity of GHGs absorbed within the Earth's oceans increases, resulting in warming ocean temperatures. As the Earth's oceans continue to warm, "global hurricane risk is expected to rise" (Balaguru et al., 2023). Hurricanes directly rely on warm water environments to develop, extracting heat energy from water sources to supply and exacerbate storm conditions. As additional heat is trapped within the Earth's oceans, hurricanes are provided with a surplus of energy to develop, leading to the creation of catastrophic storm conditions and their subsequent

impacts. Currently, hurricanes “rank among the leading causes of economic damages in the United States annually with wide-ranging societal impacts upon landfall,” and could worsen as the climate continues to warm (Balaguru et al., 2023). Beyond economic impacts, hurricanes can have a variety of lasting impacts on vulnerable ecological and social communities.

Several Ecological Impacts of Amplified Hurricane Conditions

Hurricanes can directly influence “shoreline and dune erosion” rates, directly impacting “shoreline vegetation community structure” (Greening et al., 2006). For reference, as hurricanes pass near shorelines, sediments from the shore are eroded into the ocean. Furthermore, as the strength of hurricanes are amplified by climate change, the rates at which shorelines erode hastens. As shorelines diminish, vegetation, such as mangroves and other coastal species, may lose their attachment to shore sediments, resulting in mass “defoliation” (Greening et al., 2006). These vegetative species are essential for preventing long-term shoreline erosion through containing loose sediments within their root structure. Without these species, shoreline structure may become far more susceptible to long term erosion and disturbance, impacting shore’s physical appearance and internal community function.

Moreover, hurricane-related storm surge and flooding can lead to “excessive nutrient loading[,] algal blooms[,] hypoxia[, and thus] fish and invertebrate kills” (Greening et al., 2006). When hurricanes pass over a landscape, a copious amount of rain is typically dispersed over the land’s surface. As rain makes contact with the surface and flows across the landscape, it has the possibility to absorb chemical treatments applied to the surface. Once rain is dispersed within larger water ways, the chemical treatments that were absorbed by the rain tend to flow within water ways until they eventually accumulate within larger water bodies. For reference, chemicals, such as fertilizer treatments, can contain high levels of nutrients that can be consumed

by harmful species such as algae. Once algae consumes excess nutrients, the algae can grow to the point of overpopulating water bodies, resulting in an algae bloom. Once an algae bloom dies, and is consumed by other species such as zooplankton, any leftover organic waste produced can fall to the bottom of the water body. Accumulated waste can then draw oxygen towards the water body's base, resulting in hypoxia (National Oceanic and Atmospheric Administration, 2015). Hypoxic environments contain oxygen levels that are "too low to support most life," resulting in "fish and invertebrate kills" (Greening et al., 2006; National Oceanic and Atmospheric Administration, 2015). Greening et al. provides a variety of other hurricane-related impacts felt by coastal ecological communities: "aquatic animal displacements; large scale releases of chemical pollutants and debris; exacerbated spread of exotic species and pathogens; and pollution of water with fecal microbial pathogens[, etc.]" (2006). Beyond ecological impacts, exacerbated hurricane conditions can directly impact social communities that reside near coastlines and large water bodies.

Several Social Impacts of Amplified Hurricane Conditions

As of 2003, "approximately 53% of the US population, or 153 million people, lived in coastal counties" (Bjarnadottir et al., 2011). Due to their vicinity to large water bodies, these individuals may be exposed to increasingly devastating hurricane conditions as the climate changes. These impacts have been documented over various years, including in 2004 to 2005 the "Atlantic hurricane season [was] estimated to have caused $150 billion dollars in damages [alone]" (Bjarnadottir et al., 2011). Unfortunately, some individuals, or entire communities, may be unable to pay for the recovery of infrastructure or other assets, resulting in long-term asset degradation. For example, previously economically, physically, or socially vulnerable individuals may be unable to afford the reconstruction of damaged property, resulting in the build up of poor

infrastructural conditions (Bjarnadottir et al., 2011). Thus, these poor conditions may place these individuals at more risk for eventual injury and harm. This environmental injustice, among other social impacts of amplified hurricanes, highlights the need for community planning that considers the concepts of sustainability and resilience.

Acknowledging Sustainability and Resilience Principles in Planning

In order to mitigate possible damages from intensified storms, proper planning, which considers sustainability and resilience principles, is key. Sustainable development can be defined as development that meets "the needs of the present without compromising the ability of future generations to meet their own needs" (United Nations, n.d.-b). Thus, sustainable planning can be interpreted as planning that serves to protect the needs of current communities while acknowledging the subsequent impacts of these decisions on future generations. Utilizing this logic, planners should consider the long-term benefits of sustainable designs when faced with the short-term costs of implementation. In this way, planners not only benefit present communities, but also safeguard the needs of future generations through sustainable community development.

Planners should also consider the long term resiliency of their decisions before development transpires. Resilience can be defined as "an ability to recover from or adjust easily to misfortune or change" ("Resilience," 2025). Thus, resilient planning considers potential hazards, adopting plans that account for conceivable disaster and resulting damages. Additionally, resilient planning is adaptable in that it is flexible to change with altered conditions. As coastal communities continue to bear the worsening impacts of hurricanes, planners should consider composing sustainable and resilient decisions that have the ability to mitigate disturbance. One way to begin understanding the development of sustainable and resilient plans is through referencing other community's responses to similar adversity.

Utilizing Contemporary Planning Methods to Inform Decisions

Various coastal communities across the United States have faced intensifying hurricane conditions, leading to social, economic, and ecological disturbance. For these reasons, numerous planners across the nation have adjusted their planning styles to enhance the sustainability and resiliency of their communities. Within October of 2012, Hurricane Sandy made landfall on New York Harbor, specifically within Staten Island's eastmost Raritan Bay. For reference, the Raritan Bay area "is a shallow estuary that has historically supported commercial fisheries and shell fisheries" throughout Staten Island's history (Palinkas et al., 2022). Due to Hurricane Sandy's intensity, numerous lives, homes, and businesses were lost within the bay area. Beyond these extensive impacts, Hurricane Sandy also left behind countless economic and ecological devastations in its wake.

To begin rebuilding what they had lost, the local community held a design competition titled "Rebuild by Design," which sought innovative and sustainable coastal designs from a variety of planners and designers (Palinkas et al., 2022). After careful selection, the "Living Breakwaters" project was selected for future development. This project consisted of "approximately 2,500 linear feet (~760 m) of nearshore 'breakwaters,' or partially submerged rubble-mound structures located between 790 and 1,800 ft (~240 and 550 m, respectively) from shore" (Palinkas et al., 2022). This project primarily aimed to reduce long term coastal erosion and wave run-up that had previously depreciated the area's shorelines.

To address the long-term effects of this project on local communities, a variety of other initializations were undertaken to promote sustainable community development. For example, "ecological enhancement" was a major focus of this particular project. Specifically, it concentrated on supporting ecological communities through diversifying the physical structure of

coastal habitats. This was completed through the addition of “concrete armor units that provide textured surfaces” and “‘reef streets,’ [or] narrowly-spaced rocky protrusions on the ocean side of the breakwaters” (Palinkas et al., 2022). Beyond this, planners also considered stakeholder interests through engaging with the bay’s local “Citizens Advisory Committee (CAC)” (Palinkas et al., 2022). During the project’s duration, the CAC served “a community-based advisory role to the project while leaving additional input from the public during public engagements and workshops” (Palinkas et al., 2022). Through these processes, those directly, or indirectly, involved in the project's development were able to adjust plans to best fit the community’s current structure. Thus, the Living Breakwaters project depicts that resilient and sustainable coastal solutions can be feasibly implemented within unique locations, such as within the Raritan Bay area. Although these cases are essential for planners to consider for the development of their own plans, it is also crucial to understand that unique community structures require specified plans that may face unique constraints.

Understanding Unique Communities and Constraints

Through researching contemporary mitigation plans, planners can inform decisions considering their own community. However, it is essential to acknowledge the vital differences between the communities studied and the planner’s current area of interest. For example, although communities within the Raritan Bay area have faced similar hurricane-related impacts to other coastal communities, the solutions they devised may not be feasible within differently structured communities. For these reasons, it is imperative that decisions are tailored to unique community structures, acknowledging the varied constraints that may arise during the planning process. Through online software, such as Geographic Information Systems (GIS), and direct

community engagement, plans can be modified to meet the needs of unique ecological and social communities.

Geographic Information Systems, also known as GIS, is “just one of the formalised computer-based information systems capable of integrating data from various sources to provide the information necessary for effective decision-making in urban planning” (Yeh, 1999). Through the use of GIS, planners are able to store and incorporate data from a variety of credible sources to spatially analyze social, ecological, and economic patterns, among other aspects. GIS can also be utilized for the “prediction and projection” of community structure within the future (Yeh, 1999). For example, through GIS applications, a planner can analyze how the development of a new coastal project may impact the community’s sustainability and resilience to erosion long term. If long term predictions appear to depict a low impact to protected coastlines, the planner may be inclined to push the implementation of the plan to enhance the community’s long term growth. Lastly, “when used together with remote sensing,” GIS can be utilized far after plans are fully implemented to monitor how plans have impacted community structure, allowing planners to adapt plans to further fit the community’s needs (Yeh, 1999). Thus, software, such as GIS, can allow planners to interpret the patterns within their community, tailoring plans to match these insights.

Planners should also seek out direct engagement with their community, acknowledging stakeholder opinions throughout the planning process. Engagement can be facilitated through a variety of methods. For example, planners can engage with stakeholders directly in person, or via online methods, such as through digital surveys. Through these engagements, planners can come to realize possible blind spots in their interpretations, allowing the community and its stakeholders to tailor plans to fit their needs. Similar to GIS, community engagement can occur

far after the implementation stage, offering an indicator to establish if plans may or may not be meeting necessary goals.

Conclusion

In conclusion, through proper mitigative planning, planners can offer protection to ecological and social coastal communities in the face of intensifying hurricane conditions. With society's continued use of nonrenewable resources, GHGs will continue to accumulate in the atmosphere. As these gasses accumulate, and garner excessive heat, climate conditions will change, influencing the severity of major weather events, such as hurricanes. Unfortunately, as hurricane conditions amplify, social and ecological coastal communities will be placed at more risk. For these reasons, planners must consider implementing sustainable and resilient measures within existing communities. These plans can be directly informed through researching communities that have faced similar adversity, and the solutions they proposed to mitigate disturbance. It is also essential for planners to take into account the uniqueness and constraints of their own communities when devising plans. These unique conditions can be identified through a variety of methods. For reference, spatial analysis, as performed through GIS, can depict current and future spatial patterns, impacting the feasibility of plan implementation. Furthermore, community engagement can inform planners on how to incorporate the personal opinions of stakeholders within their decisions. In the end, planning can help to shape a community's future, preserving the safety and wellness of all of those that reside within it.

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